Midterm Assignment-02

Q.1

a. The last four digit of my student id is 0532. Now, I pick last 3 digit for calculation.

```
int A[4]={0};
int i,n;
n=532;
```

Now, for(i=0; i<4; i++)

i=0 ;   i<4 ;   ~~i++~~

0<4 — True

A[0] = 532+0
= 532

if (532%2!=0) — false

i++ → i=1

i=1 ,   1<4 — True

A[1] = 532+1
= 533

if(533%2!=0) — True

A[1] = A[1]* 2
= 533 * 2
= 1066

i++ → i=2

i = 2 .

i < 4

2 < 4 — True

A[2] = 532 + 2

= 534

if (534 % 2 != 0) — false

i++ → i = 3

i = 3

i < 4

3 < 4 — True

A[3] = 532 + 3

= 535

if (535 % 2 != 0) — True

A[3] = A[3] * 2

= 535 * 2

= 1070

i++ ⇒ i = 4

i = 4

4 < 4 — false

Storage

A[0] = 532

A[1] = 1066

A[2] = 534

A[3] = 1070

Q.1

b.

```
int A[4] = {0};
int i, n;
 n = 532;
i = 0;
do{
      A[i] = n+i;
      if(A[i]%2!=0){
        A[i]*=2;
      }
      i++;
   }
while(i<4);
}
```

Q.2

my student id last four digit is 0532. Now I pick last 3 digit for calculation, which is 532. And last 2 digit is 32

```
#include<stdio.h>
int main()
{
    int num1=32, num2=532; // last 2 & 3 digit
    int b = (32%21)+5;
    int A[10];
    int a = num2;
for(int i=0; i<10; i++){
        if(i%2==0)
    { A[i]=(a%7)+3*i;
    }
   int sum=0;
   for(int i=0; i<10; i++){
        if(i%2==0){
            sum+=A[i];
        }
   }
 return 0;
}
```

Q.3

my student Id last 2 digit is: 32

that means, $a = (32 \% 3) + 2$;

$a = 4$;

~~Now~~, Here,

int arr[5][5], i, j, t1 = 0, t2 = 1, t3, x, y, z;

Now,

$i = 0$, $i < 4$ — True

t1 = 0,
t2 = 1,
x = 0,
y = 1,
z = 1,

j = 0, 0 < 4-1, 0 < 3 — True

t3 = 1,

arr[0][0] = 1

t1 = 1

t2 = 1

j = 1, 1 < 3 — True

t3 = 2

arr[1][0] = 2

t1 = 1

t2 = 2

j = 2, 2 < 3 — True

t3 = 3

arr[2][0] = 3

t1 = 2

t2 = 3

After ~~outer~~ inner loop

j = 3 3 < 3 — false

i = 1 ;

t1 = y, t1 = 1

t2 = z, t2 = 1

i=1, 1<4 — True

x=1, t1=1,
y=1, t2=1,
z=2,

j=0, 0<3 — True

t3 = 2
arr [0] [1] = 2
t1 = 1;
t2 = 2;

j= 1; 1<3 — True

t3 = 3
arr[1] [1] = 3
t1 = 2,
t2 = 3,

j=2, 2<3 — Tree

t3=5
arr[2][1] = 5,

t1=3
t2=5,

After inner loop, j=3 3<3 — false

i=2, 2<4 – True

t1=y => 1,
t2=z => 2,

x=1,
y=2,
z=3,

j=0, 0<3 — True

t3 = 3,
arr [0] [2] = 3,
t1 = 2,
t2 = 3,

j=1; 1<3 => True

t3 = 5,
arr[1] [2] = 5;
t1 = 3,
t2 = 5,

j=2, 2<3 — True

+3 = 8

arr[2][2] = 8

+1 = 5

+2 = 8

j=3 3<3 — false

After inner loop

i=3, i<4 — True,

j=0, 0<3 — True

+1 = y => 2

+3=5,

+2= z => 3,

arr[0][3] = 5,

x = 2

+1 = 3 ;

+2 = 5 ;

y = 3

z = 5

j=1, +3 = 8

arr [1] [3] = 8.

+1 = 5,

+2 = 8

j=2, +3 = 13

arr [2] [3] = 13

+1 = 8,

+2 = 13

j=3 3<3 — false

+1 = 3,

+2 = 5 ,

i=4, i<4

4<4 — false

output ~~format~~, format,

1	2	3	5
2	3	5	8
3	5	8	13